Derivative Securities FNCE 30007

Tutorial 10

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Review

Interest Rate Swap

- Interest rate swaps involve two counterparties exchanging one type of debt obligations with another type of debt obligations in the same currency.
- A standard vanilla interest rate swap is a fixed for floating swap.
- An interest rate swap is feasible if there exists a quality spread differential (QSD). We will see this in this tutorial.
- In reality, a swap bank is normally involved in the actual swap transactions.
- The principal in an interest rate swap does not change hands.

Review

Steps to design a swap

1. Calculate the QSD (also, find the comparative advantage).
2. From the benefit sharing plan, pin down the net benefit for each party. Usually, both parties equally share the benefit.
3. Find out the rates between the swap parties by matching the all-in-cost.

The third step does not have a unique answer, there may be infinite pairs of rates leading to the same net all-in-cost.

Review

Valuation of Interest Rate Swaps

- When a swap is first entered into, it has a value of 0 or close to 0. As time change, the swap will have value.
- We will value interest rate swaps as the difference between the value of a fixed-rate bond and the value of a floating-rate bond.
  - The floating-rate bond is valued by noting that it is worth par immediately after the next payment date.
  - Suppose now it is time 0, the next payment date is t, suppose the next interest payment is k^* and the notional principal is L, the floating rate is r^* (the holder already known this rate for the next interest payment). Then the value of the floating-rate bond is $(L + k^*)e^{-r^*t^*}$.
- When fixed CFs are received and floating CFs are paid, this can be regarded as a long position in fixed rate bond and a short position in floating rate bond. The value is $V_{swap} = B_{fix} - B_{float}$.

Tutorial Question

Q1: Suppose that zero interest rates with continuous compounding are as follows:

Maturity (years)	**Zero rate** r_n (% p.a.)
1	2.0%
2	3.0%
3	3.7%
4	4.2%
5	4.5%

Calculate the forward interest rates $f_{n-1,n}$ for the second, third, fourth, and fifth years, assuming continuous compounding.

Forward Rates

Under continuous compounding, the relationship between the T_x, T_y-year zero rate R_x, R_y and the $T_y - T_x$ forward rate f_{T_x,T_y} that applies between $t = T_x$ and $t = T_y$ is

$$e^{R_y * T_y} = e^{R_x * T_x} \times e^{f_{T_x,T_y} * (T_y - T_x)}$$

Equivalently,

$$f_{T_x,T_y} = \frac{R_y T_y - R_x T_x}{T_y - T_x}$$

Tutorial Question

$$f_{1,2} = 2\,r_2 - 1\,r_1 = 2 \cdot 3.0\% \; - \; 1 \cdot 2.0\% = 6.0\% - 2.0\% = 4.0\%,$$

$$f_{2,3} = 3\,r_3 - 2\,r_2 = 3 \cdot 3.7\% \; - \; 2 \cdot 3.0\% = 11.1\% - 6.0\% = 5.1\%,$$

$$f_{3,4} = 4\,r_4 - 3\,r_3 = 4 \cdot 4.2\% \; - \; 3 \cdot 3.7\% = 16.8\% - 11.1\% = 5.7\%,$$

$$f_{4,5} = 5\,r_5 - 4\,r_4 = 5 \cdot 4.5\% \; - \; 4 \cdot 4.2\% = 22.5\% - 16.8\% = 5.7\%.$$

Tutorial Question

Q2: Explain the difference between the credit risk and market risk in a financial contract.

Credit risk arises from the possibility of a default by the counter party. Market risk arises from movements in market variables such as interest rates and exchange rates. Credit risk in a swap in contingent on the values of the market variables, however a company's position in a swap only has credit risk when the value of the swap to the company is positive.

Tutorial Question

Companies X and Y have been offered the following rates per annum on a USD 5 million, 10-year investment:

	Fixed Rate	**Floating Rate**
Company X	8.0%	SOFR
Company Y	8.8%	SOFR

Company X requires a fixed-rate investment; Company Y requires a floating-rate investment. Design an interest-rate swap, acting through an intermediary bank, that

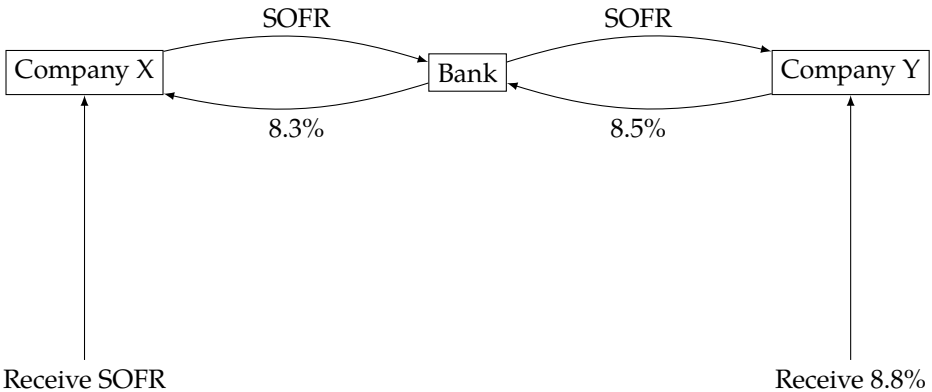
1. Intermediate bank charge a spread of 0.2% per annum, and
2. Appears equally attractive to both X and Y.

Tutorial Question

	Fixed Rate	Floating Rate
Company X	8.0%	SOFR
Company Y	8.8%	SOFR
Difference	0.8%	0

- The Maximum saving is difference in differences: 0.8% - 0 = 0.8%. (QSD)
- This means that the total apparent benefit to all parties from the swap is 0.8% p.a.
- The intermediate bank will charge 0.2% out of it.
- This leaves 0.3% p.a. better-off for each of X and Y.
- Which means X can ends up with a fixed-rate of return at 8.3%. (Note this is investment, not borrowing). Y can ends up with a floating rate of return at SOFR + 0.3%.

Tutorial Question



Tutorial Question

Q4: A financial institution has entered into an interest rate swap with company X. Under the terms of the swap, it receives 4% per annum and pays six-month SOFR on a principal of $10 million for five years. Payments are made every six months.

Suppose that company X defaults on the sixth payment date (end of year 3) when the six-month forward SOFR rates for all maturities are 2% per annum.

What is the loss to the financial institution?

Assume that six-month SOFR was 3% per annum halfway through year 3 and that, at the time of default, all zero rates are 1.8% per annum. Zero rates are expressed with continuous compounding; other rates are expressed with semiannual compounding.

Tutorial Question

Before calculating the loss, we need to identify **which interest rate applies to which cash flow**.

Rate	Used for	Why?
4%	Fixed payments	The institution receives a fixed rate of 4% throughout the swap.
3%	Floating payment at year 3	SOFR was fixed at year 2.5 for the payment due at year 3.
2%	Future floating payments	Forward SOFR represents the floating rates expected for the remaining periods.
1.8%	Discounting	The continuously compounded zero rate is used to value future cash flows at year 3.

Key idea: SOFR rates determine the floating cash flows, while zero rates determine their present values.

Tutorial Question

Company X defaults at the **end of year 3**. The swap has two years remaining.

1. Payment due at year 3
The floating rate was already fixed at year 2.5. Therefore, we use the **3% SOFR**

$$\begin{aligned}\text{Net payment at year 3} &= 10 \times 0.5 \times (4\% - 3\%) \\ &= 0.05 \text{ million.}\end{aligned}$$

2. Payments after year 3
The future floating rates are not yet realised. We use the **2% forward SOFR**.

$$\begin{aligned}\text{Net payment per period} &= 10 \times 0.5 \times (4\% - 2\%) \\ &= 0.10 \text{ million.}\end{aligned}$$

There are four remaining payments at years 3.5, 4, 4.5 and 5.

3. Valuation at year 3
Discount the four future payments using the **1.8% continuously compounded zero rate**:

$$PV = \sum_{i=1}^{4} CF_i e^{-0.018 t_i}, \qquad t_i = 0.5,\ 1,\ 1.5,\ 2.$$

Tutorial Question

The loss is the present value of the net cash flows that the financial institution will no longer receive.

Step 1: Net payment due at the end of year 3

$$\begin{aligned}\text{Fixed payment} &= 0.5 \times 4\% \times 10 = 0.20 \text{ mil},\\ \text{Floating payment} &= 0.5 \times 3\% \times 10 = 0.15 \text{ mil},\\ \text{Immediate loss} &= 0.20 - 0.15 = 0.05 \text{ mil}.\end{aligned}$$

Step 2: Value of the remaining swap payments
Four payments remain at years 3.5, 4, 4.5 and 5. Using the 2% forward SOFR:

$$\begin{aligned}\text{Fixed payment} &= 0.20 \text{ mil},\\ \text{Floating payment} &= 0.5 \times 2\% \times 10 = 0.10 \text{ mil},\\ \text{Net payment} &= 0.20 - 0.10 = 0.10 \text{ mil}.\end{aligned}$$

Step 3: Discount cash flows to the default date
Using the continuously compounded zero rate of 1.8%:

$$\begin{aligned} L &= 0.05 + 0.10e^{-0.018(0.5)} + 0.10e^{-0.018(1)} \\ &\quad + 0.10e^{-0.018(1.5)} + 0.10e^{-0.018(2)} \\ &= 0.441120 \text{ mil} \\ &= \boxed{\$441,120}. \end{aligned}$$

Tutorial Question

Q5: Six-month SOFR is 5%. SOFR forward rates for the 6–12 month period and the 12–18 month period are both 5.5%. Fixed rates for the 2- and 3-year semiannual pay swaps are 5.4% and 5.6%, respectively. Estimate the SOFR forward rates for maturities of 18 months to 2 years, 2 to 2.5 years, and 2.5 to 3 years. Assume that the 2.5-year fixed swap rate is the average of the 2- and 3-year fixed swap rates.

The continuously compounded zero rates are:

Maturity	Zero Rate
0.5 year	4.20%
1.0 year	4.35%
1.5 years	4.50%
2.0 years	4.65%
2.5 years	4.80%
3.0 years	4.95%

All zero rates are continuously compounded; all other rates are semiannually compounded.

Tutorial Question – Understanding the Rates

We are given three types of interest rates:

1. **SOFR forward rates: Floating payments**
   - 0–6 months: 5% (current six-month SOFR)
   - 6–12 and 12–18 months: 5.5%
   - 18–24, 24–30 and 30–36 months: unknown
2. **Swap rates: Fixed payments**
   - 2-year swap: 5.4%
   - 2.5-year swap: 5.5%
   - 3-year swap: 5.6%

3. Zero rates: Discounting cash flows
The discount factor for a payment at time t is

$$D(t) = e^{-R(t)t}.$$

Key principle: At inception, a par swap has zero value. Hence,

$$\boxed{\sum_{i=1}^{n} 0.5\,(F_i - K)\,D(t_i) = 0}$$

where F_i is forward SOFR and K is the fixed swap rate.

Step 1: Find the 18-month to 2-year forward SOFR

The two-year fixed swap rate is 5.4%. Let F_4 denote the unknown forward rate for the 18-month to 2-year period.

Setting the value of the swap equal to zero:

$$\begin{aligned} 0 = {} & 0.5(0.05 - 0.054)e^{-0.042(0.5)} \\ & + 0.5(0.055 - 0.054)e^{-0.0435(1)} \\ & + 0.5(0.055 - 0.054)e^{-0.045(1.5)} \\ & + 0.5(F_4 - 0.054)e^{-0.0465(2)}. \end{aligned}$$

Therefore, the 18-month to 2-year forward SOFR is

$$\boxed{F_4 = 5.62\%}$$

Step 2: Find the 2-year to 2.5-year forward SOFR

The 2.5-year fixed swap rate is

$$K_{2.5} = \frac{5.4\% + 5.6\%}{2} = 5.5\%.$$

Let F_5 denote the unknown forward SOFR for the 2-year to 2.5-year period.

Setting the value of the swap equal to zero:

$$\begin{aligned} 0 = {} & 0.5(0.05 - 0.055)e^{-0.042(0.5)} \\ & + 0.5(0.055 - 0.055)e^{-0.0435(1)} \\ & + 0.5(0.055 - 0.055)e^{-0.045(1.5)} \\ & + 0.5(0.05622 - 0.055)e^{-0.0465(2)} \\ & + 0.5(F_5 - 0.055)e^{-0.048(2.5)}. \end{aligned}$$

Solving for F_5: $\boxed{F_5 = 5.93\%}$. Similarity, $\boxed{F_6 = 6.14\%}$.